



Assessment Submission Form

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Assessment Title	Hotel Booking Management System
Module Code	B103R
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Github Link	https://github.com/pratikshingala03/HotelBbooking-System
YouTube link	https://youtu.be/fJG69qLTRDM

Declaration of Authorship:

By submitting this assignment, I declare that all material in this assessment is my own work except where there is clear acknowledgement and appropriate reference to the work of others.

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Hotel Booking Management System

1. Introduction

This project presents the design and implementation of a relational database system for a Hotel Management System using SQL. The main purpose of the system is to store and manage hotel data in a structured way, so that common hotel operations can be handled efficiently and accurately.

I selected the hotel domain because it includes meaningful real-world relationships between entities such as guests, rooms, room types, and bookings. It is also a good example for applying core database concepts taught in this module, including normalization, primary and foreign keys, constraints, indexing, and practical SQL queries.

The project objectives are:

- To design a clear and logical relational schema.
- To enforce data integrity through constraints..
- To implement useful analytical queries for business insight.

The implemented system focuses on booking management and room allocation. It allows hotel records to be created, retrieved, updated, and deleted while maintaining referential integrity between related tables.

2. System Design

2.1 Domain and Entities

The system contains four main entities:

1. **room types**
Stores room category information, for example Single, Double, Deluxe Suite, Penthouse, and each category's price per night.
2. **rooms**
Stores individual rooms with room number, room type, and current status (Available, Occupied, Cleaning, Maintenance).
3. **guests**
Stores guest profile details such as name, email, phone, and record creation timestamp.
4. **bookings**
Stores booking transactions linking a guest to a room with check-in and check-out dates, total price, and booking status.

2.2 Relationships

- One **room_type** can be linked to many **rooms** (1:M).
- One **guest** can have many **bookings** (1:M).

- One **room** can appear in many **bookings** over time (1:M).

These relationships are implemented using foreign keys:

- `rooms.type_id` references `room_types.type_id`
- `bookings.guest_id` references `guests.guest_id`
- `bookings.room_id` references `rooms.room_id`

3. Implementation

3.1 Database and Table Creation

The database `hotel_db` is created first, followed by table definitions for the four entities. Each table uses an integer auto-increment primary key for stable unique identification.

3.2 Constraints and Data Integrity

To ensure high data quality, the following constraints are used:

- Primary keys on all major tables.
- Foreign keys to preserve table relationships.
- NOT NULL for required fields.
- UNIQUE on `room_number`, `type_name`, and `email`.

4. Results

This section explains how the final system meets the project goals.

1. Basic Verification

After running the schema script, `SHOW TABLES`; confirms all four core tables are successfully created. A join query combining `bookings`, `guests`, `rooms`, and `room types` validates the relational design and confirms that foreign key relationships are correct.

2. Alignment with Project Goals

The system meets the assignment scope because it:

- stores structured relational data,
- enforces constraints and referential integrity,
- supports end-to-end data interaction through SQL,

- produces useful outputs through joins and aggregates.

5. Challenges and Solutions

During implementation, a few practical issues appeared.

Challenge 1: Ensuring data consistency in bookings

A common risk in booking systems is entering invalid date ranges or disconnected foreign key values.

Solution:

I applied strict checks (`check_out > check_in`) and foreign key constraints to block inconsistent booking records.

Challenge 2: Query performance considerations

As data grows, frequent searches on bookings and status can become slower.

Solution:

I introduced indexes on key lookup columns and date columns used in filtering and joins.

6. Conclusion and Future Work

This project successfully implements a relational database for hotel management using SQL. The final system demonstrates:

- clear schema design,
- normalized structure,
- constraints for integrity,
- indexing for optimization,

The project helped apply module concepts in a practical and structured way. It also showed how database design decisions affect reliability, maintainability, and query efficiency.

For future development, the system can be expanded with:

- payment and invoice tables,
- staff and role management,
- seasonal/dynamic pricing logic,
- room service and housekeeping task logs,
- auditing features (history of status changes),

Overall, the project achieves its intended objectives and provides a strong foundation for a larger hotel information system.

7. References

- Oracle (n.d.) *MySQL 8.0 Reference Manual*. Available at: <https://dev.mysql.com/doc/>
- GitHub (2026) *HotelBbooking-System repository*. Available at: <https://github.com/pratikshingala03/HotelBbooking-System>